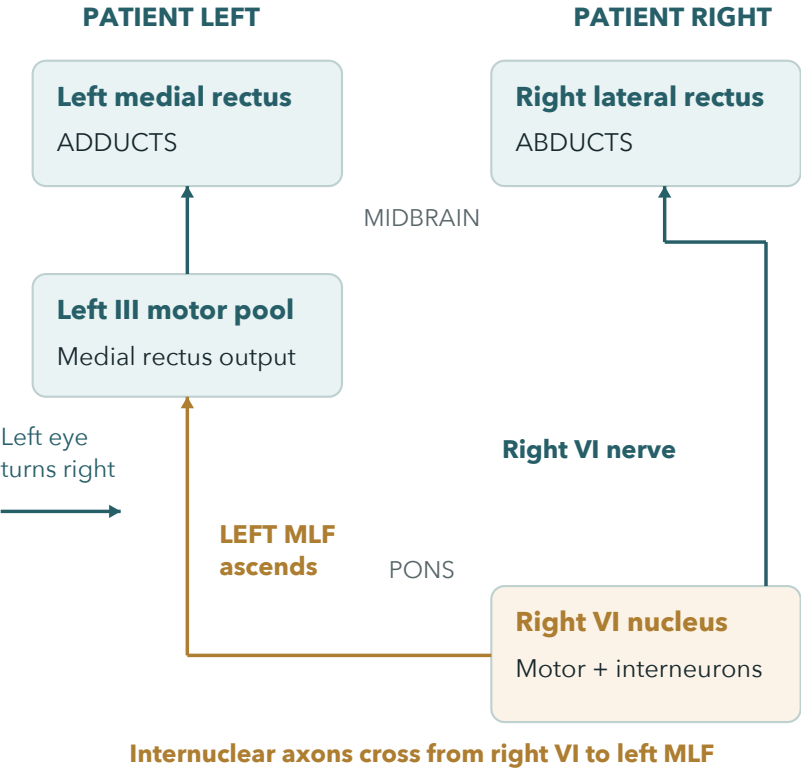


Horizontal gaze: VI and MLF

The nucleus coordinates a pair; the nerve reaches one muscle; the MLF connects the two sides.

Build one rightward gaze circuit

Right gaze uses two outputs



Functional circuit, with patient sides explicitly labelled. VI interneurons cross before ascending in the opposite MLF.

From nucleus to both eyes

Output from right VI nucleus	Effect
Motor neurons via right VI nerve	Right lateral rectus contracts: right eye abducts.
Crossed interneurons via left MLF	Left III medial rectus pool activates: left eye adducts.

Convergence is useful evidence

Near convergence can activate both medial recti through pathways distinct from this conjugate-gaze connection. Preserved convergence helps distinguish an MLF disconnection from peripheral medial rectus weakness.

Three lesions, three movement patterns

Test both eyes on right gaze

RIGHTWARD GAZE IS THE TEST

RIGHT VI NERVE

Left adduction preserved
Right abduction fails

Attempt to look right

L MR

R LR

VI

RIGHT VI NUCLEUS

Left adduction fails
Right abduction fails

Attempt to look right

L MR

R LR

VI

LEFT MLF

Left adduction fails
Right abducts; nystagmus

Attempt to look right

L MR

R LR

VI

The red interruption marks the affected output or connection. Arrows show attempted rightward gaze, not a frontal photograph.

Localise the pattern

Lesion	Right-gaze result
Right peripheral VI	Right abduction fails; left adduction remains.
Right VI nucleus	Both right abduction and left adduction fail.
Left MLF	Left adduction fails; right abduction often has nystagmus.

The laterality rule

INO is named for the eye that fails to adduct, which is the side of the MLF lesion. Convergence may be preserved, but its absence does not exclude INO.